What Likelihood Means: From a Raw Product to a Bounded, Comparable Score

Warith Harchaoui

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Abstract

This note builds up, from nothing more than the logarithm and the exponential every high-school graduate already knows, a single idea used across statistics, information theory and machine learning: read likelihood not as the raw product every textbook opens with, but as an exponentiated *average* log-density, $L := \exp(\mathbb{E}[\ln p])$. That reformulation is unusual: almost no standard reference defines likelihood this way. Every notion it leans on gets its own loop here, a plain-language intuition first, a worked numeric example, then the formula: the geometric mean, Jensen's inequality, Shannon entropy, a temperature-indexed refinement borrowed from equilibrium statistical mechanics, the Legendre transform, Cramér's large deviations, and Wald's martingale. A full account of L as a statistical estimator follows, bias and variance included, together with an exact link between its maximizer and the Kullback-Leibler divergence, Fisher information and the asymptotic efficiency of maximum likelihood included. Three new experiment figures make visible results the text used to only describe: Cramér's decay rate checked against its exact Legendre-transform rate, one path of Wald's martingale crossing a sequential-test threshold, and the Monte Carlo convergence of L's bias and variance onto the delta method's formulas. Two worked instantiations, a categorical classification model and a regression model under Gaussian noise, show the same construction landing on the reciprocal of a familiar error measure in both cases: perplexity in the first, root-mean-square error in the second. Each one is extended by a real-data experiment, real code and real training curves included: a softmax classifier trained on the Palmer Archipelago penguins (Gorman et al., 2014), a linear regressor and a small multi-layer perceptron (MLP) both trained on California housing (Pace and Barry, 1997). `elbow-helper`'s own mathematics note, `ELBOW-en.tex`, builds its Gaussian model-selection criteria (BIC, mBIC, and ICL: scores that penalize a likelihood for model complexity) on top of the foundation derived here rather than re-deriving it. Citations follow the same evidence-marking convention as `ELBOW-en.tex`: each claim states whether it is a verified citation, a citation reported through a secondary source, or this note's own construction.

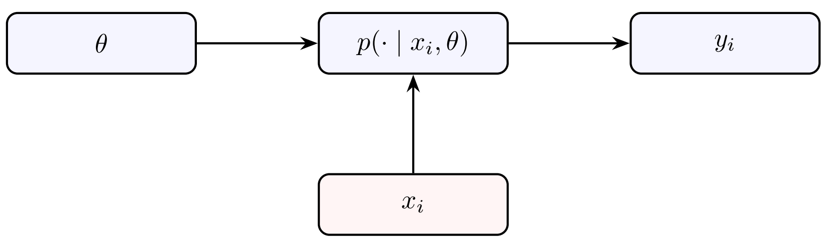
Contents

1 The notion

A statistical model answers one plain question: given what the model believes, how expected was the data actually observed? Making that question precise takes three pieces, each one already familiar under a different name.

The first piece is the data itself. This note calls "data" n pairs (x_i, y_i), an input x_i paired with an outcome y_i: a covariate driving a prediction, a position driving a curve's height, or nothing at all, x_i simply absent, for a model that only ever describes y_i on its own. The second piece is the parameter θ, the model's own dials, tuned to fit the data, the same role a line's slope and intercept already play in the regression every high-school statistics unit covers, just more general. The third piece is the *conditional density* $p(\cdot \mid x_i, \theta)$: a rule that, once x_i and θ are fixed, assigns every possible outcome a weight saying how plausible it would be. In the discrete case (a finite list of possible outcomes, like one class out of K) that weight is an ordinary probability: the weights are positive and add up to 1. In the continuous case (an outcome that can take any real value, like a physical measurement) "density" replaces "probability": $p(y)$ is no longer itself a probability, only a curve's height; the area under that curve between two bounds is the probability Y lands in that interval. A Gaussian (the bell curve) is the best-known example: its height is large near the mean and small far from it, without that height ever being a probability above 1. That distinction returns in Section 5, where it explains why a continuous likelihood behaves differently from a discrete one.

These three pieces assemble into a single diagram, the one every model this note uses shares: θ and x_i set the conditional density; y_i is drawn from it.



A box is a quantity, an arrow reads "feeds into": θ and x_i feed the density, which in turn produces y_i. This kind of picture, a graphical model, does nothing more than make visible who depends on whom.

The same symbol keeps recurring in slightly different clothes, so a quick reference is worth more than trying to hold it all in one linear pass.

Symbol	Meaning
n	number of observations in the sample
x_i	input attached to the i-th observation (optional)
y_i	observed outcome for the i-th observation
θ	model parameters, the dials fit to the data
$p(\cdot \mid x_i, \theta)$	density the model predicts for y_i
ln, exp	natural logarithm and its inverse function
$\mathbb{E}[\cdot]$	average weighted by the model's own probabilities
L_{raw}	raw product of the n densities, textbook likelihood
L	this note's exponentiated geometric-mean likelihood
Z_i	log-density $\ln p(y_i \mid x_i, \theta)$ of one observation
β	temperature parameter (Section 2)
$H(p)$	Shannon entropy of p, an uncertainty measured in nats

Table 1: Notation glossary. Every symbol is reintroduced at its first use; this table only exists so nothing has to be memorized on a single pass.

With these three pieces in hand, *likelihood* names the joint density of the observed outcomes, read as a function of the parameter, not the data: $L_{\text{raw}}(\theta) := \prod_{i=1}^{n} p(y_i \mid x_i, \theta)$, a plain product of n terms, one per observation. This is the definition every standard treatment opens with, Bishop's among the clearest (Bishop, 2006). It is also the exact quantity a Bayes factor or a Neyman-Pearson likelihood-ratio test compares directly.

High school already supplies the tool needed to see what breaks here. Multiplying n numbers between 0 and 1 shrinks with every extra factor: $0.3 \times 0.3 = 0.09$, $0.3^{10} \approx 6 \times 10^{-6}$, $0.3^{40} \approx 1 \times 10^{-21}$, no surprise to anyone who has already sketched x^n on $]0, 1[$ and watched it collapse toward 0 as n grows, whatever x was fixed at. L_{raw} meets exactly that fate once n passes a few dozen, since it multiplies n densities that typically sit well below 1. The number computed on 40 observations and the number computed on 4,000 observations are therefore not numbers a single threshold could ever compare: the second is crushed by an extra factor on the order of 0.3^{3960}, a number with thousands of zeros after the decimal point. The raw product carries the sample size baked into its own scale. A score whose scale shifts with the sample size it happens to be fed is not a comparable score.

This note departs from the textbook definition right here and says so plainly rather than leaving the departure implicit inside a formula. The fix reaches for an idea already on hand: the *geometric mean*. For two strictly positive numbers a and b, the geometric mean $\sqrt{ab}$ answers a different question from the arithmetic mean $\frac{a+b}{2}$: what single number, repeated twice and multiplied by itself, would reproduce the same product as $a \times b$? Taking the geometric mean of n numbers generalizes that move: it is the n-th root of their product, the one number whose n-th power reproduces that product. L_{raw} is exactly a product of n densities, so its geometric mean is computed directly,

$$L := \exp\bigl(\mathbb{E}[\ln p]\bigr) = L_{\text{raw}}^{1/n},$$

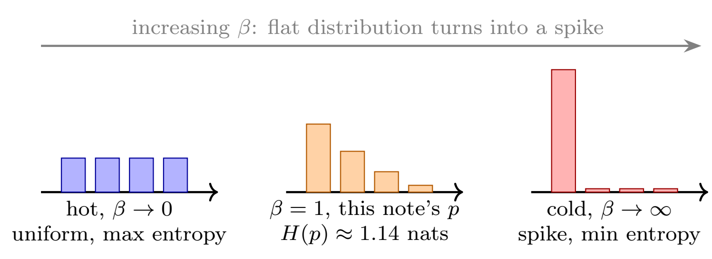
the n-th root of L_{raw}, never a different number, only a different form of the same one. The ln-and-exp notation is only a convenient detour to get there: taking the logarithm turns a product into a sum ($\ln(ab) = \ln a + \ln b$), averaging that sum divides it by n, and exponentiating back undoes the logarithm, landing on an n-th root, exactly the pair of inverse functions already handled in high-school algebra. Almost no standard reference defines likelihood this way. The reason is not stylistic: dividing the exponent's sum by n before re-exponentiating is exactly what turns an *extensive* quantity, one whose scale depends on sample size (like a total mass, which doubles if the amount of material doubles), into an *intensive* one, whose scale does not (like a density, the same whether a gram or a ton of the same material is weighed). It is the same move information theory makes when it works with *perplexity*, an exponentiated *average* log-probability rather than a raw joint one, made literal in Section 4.

2 What the number means

Before any asymptotic machinery, L has a direct reading worth sitting with. It starts from a piece of geometry already met in high school: a function is *concave* when the chord joining two points on its graph stays under the graph, never above it. ln is concave on its domain, so a chord drawn between $\ln(2)$ and $\ln(8)$ always sits below the logarithm's own curve between those two points. *Jensen's inequality* generalizes that geometric fact: for a concave function f and a random variable X, $f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$, the image of the average never falls below the average of the images. Applied to ln, then both sides re-exponentiated (an increasing function, which preserves the direction of an inequality), this gives $L = \exp(\mathbb{E}[\ln p]) \leq \mathbb{E}[p]$ always: the geometric mean this note works with never exceeds the arithmetic mean of the same densities, with equality only when p is constant across observations. A worked pair of numbers makes it concrete: for 0.2 and 0.8, the arithmetic mean is 0.5 while the geometric mean $\sqrt{0.2 \times 0.8} = 0.4$ sits below it, pulled toward the smaller of the two. That pull toward small values keeps returning through this note: a geometric mean cares more about one bad outlier than an arithmetic mean does.

Jensen's inequality is not hard to see directly. The way to see it previews the tangent-line idea returning in full geometric form in Section 3. A concave function's tangent line at any point x_0, $\ell(x) := f(x_0) + f'(x_0)(x - x_0)$, sits at or above the curve everywhere, $f(x) \leq \ell(x)$ for every x: the mirror image of a chord sitting below the curve, now for a tangent instead of a chord. Set $x_0 := \mu := \mathbb{E}[X]$, plug in the random variable X for x, and average both sides: $\mathbb{E}[f(X)] \leq \mathbb{E}[\ell(X)] = f(\mu) + f'(\mu)\big(\mathbb{E}[X] - \mu\big) = f(\mu)$, since $\mathbb{E}[X] - \mu = 0$ by definition of μ. Four lines, nothing beyond the fact that a concave curve never dips below its own tangent.

For a discrete p over K outcomes, take the expectation under p itself rather than under data: $\exp(\mathbb{E}_{y\sim p}[\ln p_y]) = \exp(-H(p))$, where $H(p) := -\sum_y p_y \ln p_y$ is p's *Shannon entropy* (MacKay, 2003), a way of pricing the uncertainty p expresses. Intuition comes before the formula here: a fair coin flip ($p = (0.5, 0.5)$) is more uncertain than a loaded die that almost always lands the same way ($p \approx (0.95, 0.05)$): $H(p)$ works out to $\ln 2 \approx 0.693$ in the first case against $H(p) \approx 0.198$ in the second, a lower number for a more predictable situation. Read through the exponential, $\exp(-H(p))$ sits in $[1/K, 1]$: equal to $1/K$ when p is uniform over the K outcomes (maximum uncertainty, no outcome favored), rising toward 1 as p concentrates on one of them (minimum uncertainty). Read this way, $\exp(-H(p))$ is a typical or effective probability. Its reciprocal $\exp(H(p))$ is the effective number of plausible outcomes: $\exp(H(p)) \approx 4$ means about as uncertain as a fair guess among four.



Shannon entropy read as one snapshot of the temperature sweep to come: each panel is the same four-category p, tilted by a different β. Flat bars (left) carry the most entropy, $\exp(-H)$ at its floor $1/K$; a single spike (right) carries the least, $\exp(-H)$ near its ceiling $\max_k p_k$. The untilted p itself (center, $\beta = 1$) sits wherever its own shape happens to fall between those two extremes.

A temperature reading. That bound has a one-parameter refinement worth making precise, not merely evocative. It reaches for an intuition already on hand: heat stirs things up, cold locks them down. For $\beta > 0$, define the power-tilted family $p_\beta(y) := p(y)^\beta / Z(\beta)$, $Z(\beta) := \int p(y)^\beta \, dy$

(or $\sum_y p(y)^\beta$ in the discrete case): a genuine exponential family (a family whose density is itself an exponential of a linear function of a parameter), with β playing the role Gibbs assigns to inverse temperature against the "energy" $-\ln p(y)$. This is exactly the object the large-deviations statistical-mechanics dictionary describes: $\ln Z(\beta)$ is a scaled free energy, β its inverse temperature. The relationship between them is a Legendre transform, the geometric tool built up in detail in Section 3 (where it directly powers Cramér's argument): it is, term for term, the same relationship that connects free energy and entropy in equilibrium statistical mechanics (Touchette, 2009). Differentiating at $\beta = 1$, where $p_\beta = p$, recovers L itself: $\frac{d}{d\beta} \ln Z(\beta)\big|_{\beta=1} = \mathbb{E}_p[\ln p] = \ln L$, so L is one derivative of a free energy, evaluated at one specific temperature. The Shannon entropy above is itself the $\beta \to 1$ limit of the Rényi entropy $H_\beta(p) := \ln Z(\beta)/(1-\beta)$, the same uncertainty measure read at every temperature β at once, collapsing to Shannon's own entropy exactly at $\beta = 1$, a fact checked directly from the definitions above.

The two extremes of β read exactly like the two extremes of temperature, made precise rather than left as image. As $\beta \to 0^+$, hot and maximally agitated, p_β flattens toward uniform over p's support regardless of what p itself looked like: $Z(\beta) \to K$ in the K-category case, so $\exp(-H_\beta(p)) \to 1/K$, exactly the floor derived above. As $\beta \to \infty$, cold and frozen, p_β collapses onto p's own mode: $\exp(-H_\beta(p)) \to \max_k p_k$, the probability of the single most likely category, a tighter, p-dependent ceiling than the loose upper bound of 1 (reached only in the degenerate limit where p itself is already a point mass). Verified numerically on $p = (0.5, 0.3, 0.15, 0.05)$: $\exp(-H_\beta(p))$ moves from $0.25 = 1/4$ at $\beta \to 0$ to $0.5 = \max_k p_k$ at $\beta \to \infty$, monotonically, passing through $\exp(-H(p)) \approx 0.32$ at $\beta = 1$. Water's own two familiar extremes at atmospheric pressure, freezing at $0\,^\circ$C and boiling at $100\,^\circ$C, are a physical illustration of exactly this shape: a fixed system swept from an ordered, low-entropy phase to a disordered, high-entropy one as an inverse-temperature-like parameter crosses its range. This is not a numerical correspondence: nothing here claims $\beta = 0$ or $\beta = \infty$ literally sit at those two Celsius values, only that the qualitative mechanism and its exact Legendre-transform bookkeeping are the ones equilibrium statistical mechanics already worked out.

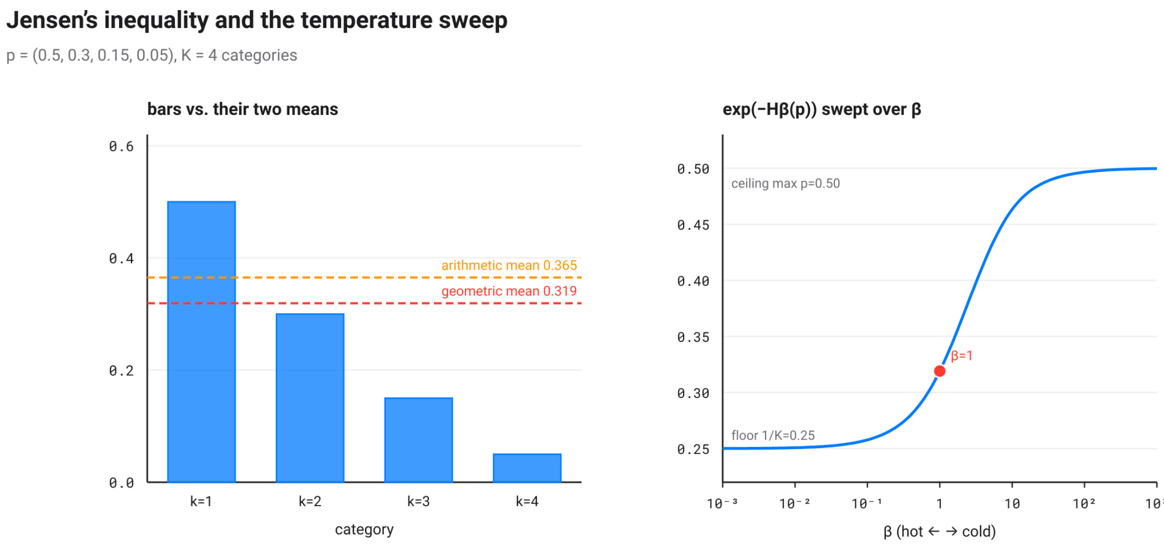
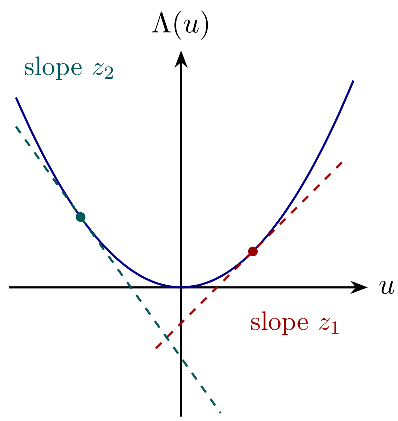


Figure 1: Left: the bar chart above, $p = (0.5, 0.3, 0.15, 0.05)$, with its geometric mean $\exp(\mathbb{E}[\ln p]) \approx 0.319$ and arithmetic mean $\mathbb{E}[p(Y)] \approx 0.365$ marked, Jensen's inequality made visible as two horizontal lines rather than left as a bare formula. Right: $\exp(-H_\beta(p))$ swept across β, hot on the left, cold on the right, moving monotonically from the floor $1/K = 0.25$ to the ceiling $\max_k p_k = 0.5$, with $\beta = 1$, this note's own L, marked where it sits along that path.

3 Why an average, not a sum: large deviations and martingales

Write $Z_i := \ln p(y_i \mid x_i, \theta)$ for the per-observation log-density, so $L = \exp\left(\frac{1}{n}\sum_i Z_i\right)$, an exponentiated sample mean. The exact relation this buys back to L_{raw} is worth spelling out: $L^n = \exp\left(\sum_i Z_i\right) = \prod_i p(y_i \mid x_i, \theta) = L_{\text{raw}}(\theta)$. Taking the n-th root and re-exponentiating an average therefore loses no information relative to the raw product: it is the same number, an n-th root away, now expressed in the one form whose large-sample behavior is understood. That behavior is partly named already by the law of large numbers: a sample mean like $\frac{1}{n}\sum_i Z_i$, computed on observations drawn independently from the same distribution, edges closer to its theoretical value $\mathbb{E}[Z]$ as n grows, the same guarantee that pushes an observed coin-flip frequency ever closer to 1/2 the more times the coin is tossed. That guarantee promises convergence; it says nothing about the speed. Two classical results fill that gap; each says, in its own way, that working with an average rather than a raw product is not merely a convenient rescaling.

A necessary detour: what is a Legendre transform? The result that comes next, Cramér's, is stated using a geometric tool that earns its own loop before being put to work. Start from a convex curve (the mirror image of concave: its chords sit above it, never below). Then ask, for every possible slope z, which line of that slope touches the curve without ever rising above it. Exactly one line of slope z fits that description: the tangent to the curve at the point where its own slope equals z. The curve's Legendre transform assigns to each slope z the vertical gap between that tangent line and the origin, flipped in sign: a way of re-encoding the whole curve, one tangent point at a time, in a form that swaps the roles of "position" and "slope."



For a given slope z, exactly one line of that slope touches the convex curve Λ without ever rising above it (red for z_1, teal for z_2): the tangent at the marked point. The Legendre transform $\Lambda^*(z)$ reads off, up to a sign, where that line crosses the vertical axis.

Cramér's theorem. With that tool in hand the result states directly. Cramér's theorem (Cramér, 1938), the founding result of large-deviations theory, says an independent and identically distributed (i.i.d.) sample mean like $\frac{1}{n}\sum_i Z_i$ does more than converge to $\mathbb{E}[Z]$: the probability that it strays from that limit by more than any fixed amount decays exponentially in n, at a rate set by a convex rate function, exactly the Legendre transform introduced above applied to Z's cumulant generating function. Working with the average first, unlike the raw product L^n it is one root away from, is what exposes that exponential structure, in place of burying it inside an already-multiplied number.

That rate does not have to stay abstract: on the categorical example $p = (0.5, 0.3, 0.15, 0.05)$ already plotted in Figure 1, it can be computed for real with the same tools as above. Write $\Lambda(u) :=$

$\ln \mathbb{E}[e^{uZ}] = \ln \sum_k p_k^{1+u}$ for the cumulant generating function of $Z = \ln p(Y)$ and $\mu_Z := \mathbb{E}[Z] \approx -1.1421$ for its mean (the same quantity as $-H(p)$ from Section 2, since $\mathbb{E}[Z] = \mathbb{E}[\ln p(Y)]$). The rate function at a target z is exactly Figure 3's construction applied to Λ: $I(z) := \sup_u [uz - \Lambda(u)]$, a one-variable optimization solved here numerically rather than left as a bare formula. At the two bounds Figure 2 plots, $z = \mu_Z + 0.1 \approx -1.0421$ and $z = \mu_Z - 0.1 \approx -1.2421$, this gives $I(\mu_Z + 0.1) \approx 0.0153$ and $I(\mu_Z - 0.1) \approx 0.0129$: since the two-sided event is dominated by whichever tail decays more slowly, it is the smaller of the two rates, $I \approx 0.0129$, that governs the total probability for large n.

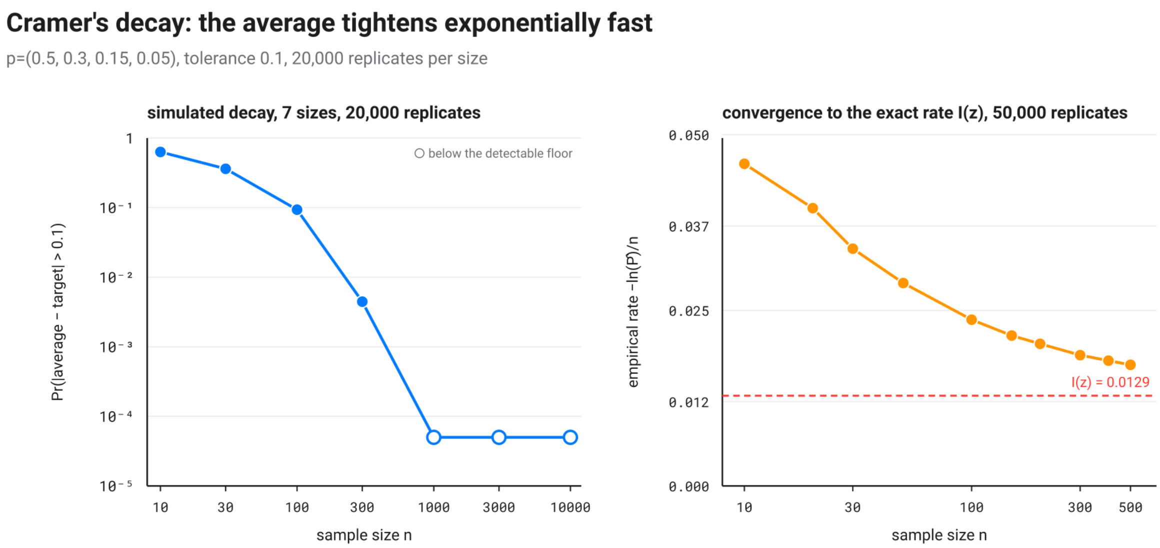


Figure 2: Monte Carlo simulation on the categorical example $p = (0.5, 0.3, 0.15, 0.05)$. Left: probability that the sample mean of $Z_i = \ln p(Y_i)$ strays from its target $\mathbb{E}[Z]$ by more than 0.1, as a function of n (20,000 replicates per size), both axes logarithmic; the near-straight path is the visual signature of exponential decay. Hollow points mark a probability that dropped to 0 across the replicates, below the detectable floor. Right: the empirical rate $-\ln(\hat{P}_n)/n$ estimated at each n (50,000 replicates, n restricted to the range where the probability stays measurable), against the exact value $I \approx 0.0129$ computed above by the Legendre transform (red line). The curve approaches it without fully arriving over this range of n: $-\ln(\hat{P}_n)/n$ converges to I only up to an $O(\ln n/n)$ correction term, the reason Cramér's result describes an asymptotic regime rather than an exact equality at finite n.

Chernoff (Chernoff, 1952) specializes exactly this machinery to a decision question, not an estimation one: how should one choose, from n i.i.d. draws, between two possible stories for the same data, two candidate densities p_0 and p_1? In the symmetric setting where both hypotheses carry a prior and a single test minimizes the total (Bayes) probability of error rather than one error type at a fixed level of the other, the best total error probability any such test can achieve decays as $\exp(-n\, C(p_0, p_1))$, where the *Chernoff information* $C(p_0, p_1) := -\min_{\lambda \in [0,1]} \ln \int p_0^{\lambda} p_1^{1-\lambda}$ is, structurally, one more exponentiated average of a log, this time an integral over the sample space, instead of a mean over observations. A number makes the object concrete: for the $p_0 = 0.3$, $p_1 = 0.7$ Bernoulli pair used below in Figure 3, symmetry places the minimizing λ at $1/2$; this gives

$C(p_0, p_1) = -\ln\left(2\sqrt{0.3 \times 0.7}\right) \approx 0.0872$, the exponential rate at which the best possible total error probability for telling the two coins apart shrinks with every extra observation. This is a different question from the one the Chernoff-Stein lemma answers, fixing one error type's probability and asking how fast the other can be driven down, which lands on the Kullback-Leibler divergence, not the Chernoff information; the two results are often confused precisely because both carry Chernoff's name.

Wald's martingale. The same running product has a second, complementary life as a *process*, not a single number. That second life answers a different question: not "how much data is needed to decide?" but "can a decision be reached as soon as the data allow, without fixing n in advance?" Define the likelihood-ratio process $\Lambda_t := \prod_{i=1}^{t} p_1(y_i)/p_0(y_i) = \exp\left(\sum_{i=1}^{t} \ln[p_1(y_i)/p_0(y_i)]\right)$ over the first t observations. A *martingale* is a process whose best prediction for tomorrow, given everything known up to today, is simply today's own value: no drift up, no drift down, the picture of a repeated fair bet. Under p_0, Λ_t is precisely that, a nonnegative martingale, $\mathbb{E}_{p_0}[\Lambda_t \mid \mathcal{F}_{t-1}] = \Lambda_{t-1}$, the fact Wald (Wald, 1945) built the *Sequential Probability Ratio Test* (SPRT) on: a threshold crossed by Λ_t at a data-dependent stopping time still controls the test's error rate, by the optional stopping theorem (a martingale stopped at a data-dependent but well-behaved stopping time still has the same expected value as at the start), the guarantee a fixed-n test gets automatically and a naively-stopped one does not.

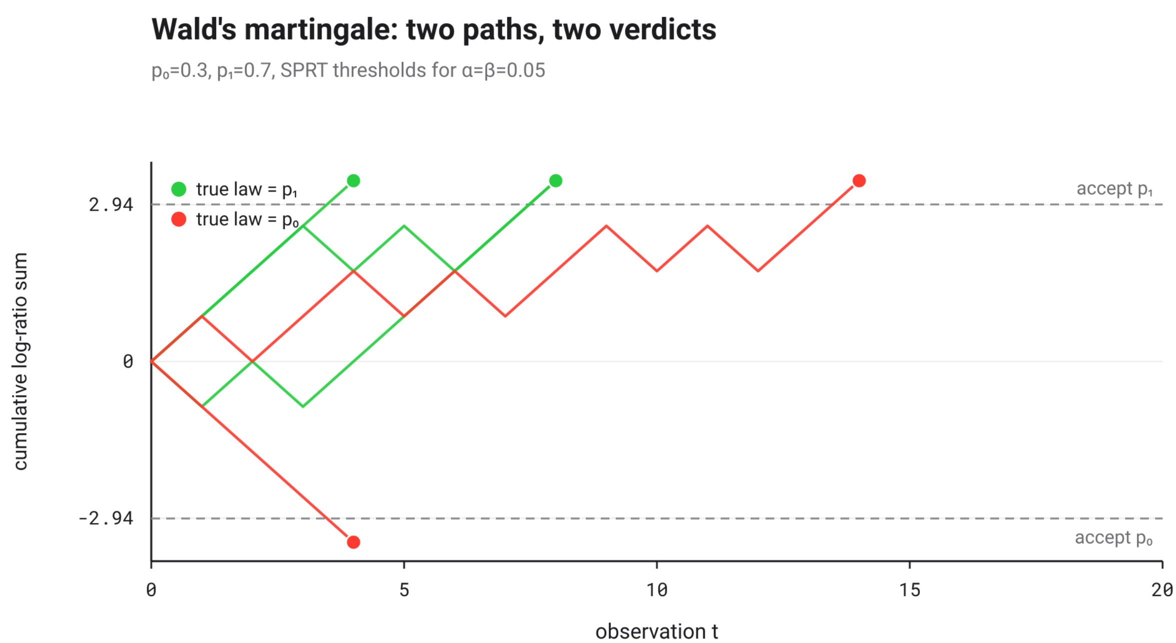


Figure 3: Simulated Λ_t on a log scale (the cumulative log-ratio sum) for $p_0 = 0.3$, $p_1 = 0.7$, with both SPRT thresholds for $\alpha = \beta = 0.05$. Green paths, simulated under the true law p_1, drift upward and eventually cross the "accept p_1" threshold; red paths, simulated under the true law p_0, drift downward and cross the other one. The test stops the moment Λ_t crosses either threshold, at a time set by the data rather than fixed in advance.

That guarantee needs two fully specified densities p_0 and p_1, though, a requirement not every sequential problem can meet; `elbow-helper`'s own sequential test (`ELBOW-en.tex`, the sequential-test section, `sec:fwer`) is exactly such a case, its alternative at each step a composite, best-fitting extra breakpoint rather than a second fixed density, which is why it calibrates against a permutation null instead of Wald's classical martingale bound.

Two different questions about L as an estimator. Everything computed so far, L included, comes from a finite sample. An *estimator* is any quantity computed from data that targets a fixed theoretical value. Two distinct questions are worth separating cleanly rather than letting either one hide inside the other. First: is the sample-based L this note computes on n real observations a good estimator of the population value it converges to as $n \to \infty$? Second: does the raw, un-rooted product L_{raw} estimate that same population value at all? Fix the target first. Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be drawn i.i.d. from the true joint data distribution q and let p_θ be a candidate model for $p(y \mid x)$. As $n \to \infty$, this note's own L is built to converge to one fixed number,

$$L_\infty := \exp(\mu_Z), \qquad \mu_Z := \mathbb{E}_q\big[\ln p_\theta(Y \mid X)\big],$$

not a random quantity: μ_Z is an ordinary expectation under the true q, a single number once p_θ and q are fixed. Two different sample-based quantities can now be compared against that one target.

The first is this note's own L, computed on n observed pairs: $\hat{L}_n := \exp(\bar{Z}_n)$, $\bar{Z}_n := \frac{1}{n}\sum_i Z_i$, $Z_i := \ln p_\theta(Y_i \mid X_i)$. $\bar{Z}_n$ is exactly unbiased for μ_Z (its average across every possible replay of the sampling lands exactly on μ_Z), so $\hat{L}_n$ is a genuine estimator of L_∞, though not an unbiased one itself: exp is convex (the mirror of ln, concave), so the same Jensen's inequality from Section 2 now runs the other way and gives $\mathbb{E}[\hat{L}_n] \geq L_\infty$ always. The size of that upward bias can be computed rather than merely asserted: the *delta method* is nothing more than a Taylor expansion applied to a random variable. Write $\varepsilon_n := \bar{Z}_n - \mu_Z$ for the random wobble around the target, with mean zero, $\mathbb{E}[\varepsilon_n] = 0$ and variance $\mathbb{E}[\varepsilon_n^2] = \sigma_Z^2/n$ (the variance of an average of n i.i.d. terms each with variance $\sigma_Z^2 := \mathrm{Var}_q[Z]$). The second-order Taylor expansion of exp around μ_Z gives $\exp(\bar{Z}_n) = \exp(\mu_Z + \varepsilon_n) = \exp(\mu_Z)\big[1 + \varepsilon_n + \varepsilon_n^2/2 + O(\varepsilon_n^3)\big]$; taking the expectation term by term gives the bias directly,

$$\mathbb{E}[\hat{L}_n] = L_\infty\Big(1 + \underbrace{\mathbb{E}[\varepsilon_n]}_{=0} + \frac{\mathbb{E}[\varepsilon_n^2]}{2} + O(n^{-3/2})\Big) = L_\infty\Big(1 + \frac{\sigma_Z^2}{2n} + O(n^{-2})\Big).$$

The variance reads off the same expansion, truncated to first order this time: $\hat{L}_n - L_\infty \approx L_\infty\, \varepsilon_n$, so

$$\mathrm{Var}[\hat{L}_n] \approx L_\infty^2\, \mathrm{Var}[\varepsilon_n] = L_\infty^2\, \frac{\sigma_Z^2}{n} + O(n^{-2}),$$

both vanishing as $n \to \infty$: $\hat{L}_n$ is consistent (its error shrinks to 0 as n grows), in fact $\sqrt{n}$-consistent (that error shrinks at the specific rate $1/\sqrt{n}$) and asymptotically normal (its rescaled error settles into a bell-curve shape): $\sqrt{n}(\hat{L}_n - L_\infty) \to \mathcal{N}(0, L_\infty^2 \sigma_Z^2)$ in distribution states that same fact formally, the rescaled error's own probability law converging to a bell curve as n grows. Its bias shrinks at the same $O(1/n)$ rate as its variance. Figure 4 checks those two formulas rather than merely asserting them: on the well-specified Gaussian case $q = p_\theta = \mathcal{N}(0, 1)$, a Monte Carlo simulation with 100,000 replicates per sample size closely matches, at every n tried, both the bias and the variance the delta method predicts.

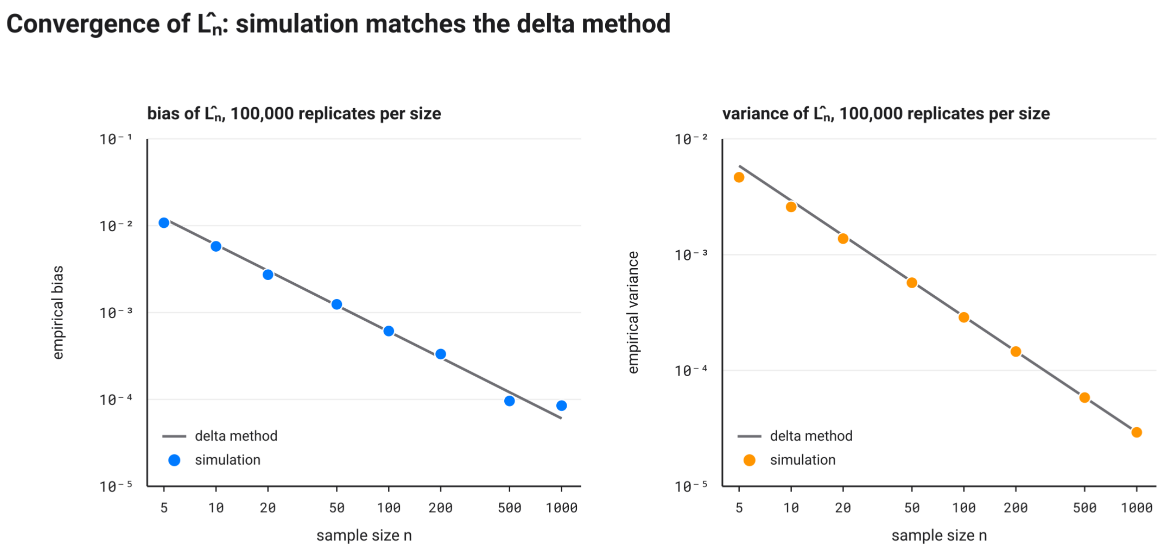


Figure 4: Well-specified case $q = p_\theta = \mathcal{N}(0,1)$, 100,000 Monte Carlo replicates per sample size n. Left: empirical bias of $\hat{L}_n$ against the delta method's prediction $L_\infty \sigma_Z^2/(2n)$. Right: empirical variance of $\hat{L}_n$ against $L_\infty^2 \sigma_Z^2/n$. Both axes are log-log; the simulated points lining up on the theoretical line is the verification, not merely the assertion, that bias and variance vanish at the announced $O(1/n)$ rate.

The second candidate is the raw product this note opened by rejecting, $L_{\text{raw}} := \prod_i p_\theta(Y_i \mid X_i) = \hat{L}_n^n = \exp(n\bar{Z}_n)$. Is it also an estimator of L_∞? Its expectation answers the question directly:

$$\mathbb{E}[L_{\text{raw}}] = \big(\mathbb{E}_q[e^Z]\big)^n = \exp\big(n\,\Lambda(1)\big), \qquad \Lambda(u) := \ln \mathbb{E}_q[e^{uZ}],$$

where Λ is exactly the cumulant generating function Cramér's argument above is built from and whose Legendre transform in Figure 3 supplies the rate function. For $\mathbb{E}[L_{\text{raw}}]$ to converge to L_∞ or to anything fixed at all, $\Lambda(1)$ would need to equal 0 exactly, a knife-edge condition with no reason to hold for a generic p_θ. Absent it, $\mathbb{E}[L_{\text{raw}}]$ decays to 0 or explodes to ∞ exponentially fast in n. L_{raw} is not a biased or noisy estimator of L_∞: it is not an estimator of it at all, the precise, quantitative version of Section 1's opening observation that L_{raw} on 40 points and on 4,000 points are not numbers a single threshold could ever compare.

Why maximizing L recovers the truth: the Kullback-Leibler divergence. Everything above fixed p_θ and read L as a score. A different, equally natural question asks the reverse: among every possible θ, which one makes $L_\infty(\theta) = \exp(\mu_Z(\theta))$ largest? The answer is an identity that reuses Jensen's inequality already established in Section 2, instead of proving a new one. Define the *Kullback-Leibler divergence* between the true distribution q and the model p_θ, $\mathrm{KL}(q\|p_\theta) := \mathbb{E}_q\big[\ln \frac{q(Y|X)}{p_\theta(Y|X)}\big]$; it is always nonnegative, a fact that follows directly from Jensen: $\mathrm{KL}(q\|p_\theta) = -\mathbb{E}_q\big[\ln \frac{p_\theta(Y|X)}{q(Y|X)}\big] \geq -\ln \mathbb{E}_q\big[\frac{p_\theta(Y|X)}{q(Y|X)}\big] = -\ln 1 = 0$, with equality exactly when $p_\theta = q$ almost everywhere. A little algebra then ties this divergence to the cross-entropy met below in Section 4.1: $-\mu_Z(\theta) = \mathrm{CE}(q, p_\theta) = H(q) + \mathrm{KL}(q\|p_\theta)$, where $H(q)$ does not depend on θ. Maximizing $L_\infty(\theta)$ over θ is therefore exactly the same as minimizing $\mathrm{KL}(q\|p_\theta)$: if the true distribution sits inside the model family, $q = p_{\theta_0}$ for some θ_0, that θ_0 is the unique maximizer, since KL vanishes only there. Maximizing L, or equivalently the classical L_{raw}, which it is one n-th root away from, is therefore not an arbitrary convention: in this precise sense it pulls the model as close to the truth as the available information allows.

Fisher information and the asymptotic efficiency of maximum likelihood. One last piece connects this result to the quality of $\hat{\theta}$ itself, the point where L reaches its maximum, not only to the quality of L as a score. An observation's *score* is the derivative $s(\theta; y) := \frac{\partial}{\partial\theta} \ln p_\theta(y)$, the log-density's sensitivity to θ; its expectation under the true θ is zero by construction. The *Fisher information* $I(\theta) := \mathrm{Var}_\theta[s(\theta; Y)] = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2} \ln p_\theta(Y)\right]$ (under standard regularity conditions) measures how sharply the log-likelihood curves around θ: high Fisher information means a tightly peaked curve, easy to pin down precisely; low information means a flat curve and a θ that is hard to localize. The Cramér-Rao bound says no unbiased estimator of θ built from n i.i.d. observations can do better than $\mathrm{Var}(\hat{\theta}) \geq 1/(nI(\theta))$, whichever estimation method is used. L's maximizer, exactly the classical maximum likelihood estimate since $L = L_{\text{raw}}^{1/n}$ shares its $\arg\max$, reaches that bound asymptotically under the same regularity conditions: $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta) \to \mathcal{N}(0, I(\theta)^{-1})$ in distribution (Bishop, 2006). Maximizing L does not just give the θ closest to the truth in the Kullback-Leibler sense: for large n, it also gives, among every unbiased estimator imaginable, about as precise a one as any method could ever reach.

Five results, one table. This section derived five different asymptotic facts about the same running average, each answering its own question about what happens as n grows. Table 2 collects them in one place before Section 4 moves on to concrete models.

Result	Answers	Rate or bound	Figure
Cramér (large deviations)	how fast the sample mean concentrates around its target	$\Pr(\lvert\bar{Z}_n - \mathbb{E}[Z]\rvert > \epsilon) \sim e^{-nI(\epsilon)}$, I the Legendre transform of Λ	Fig. 2
Chernoff information	the best total error rate for choosing between two densities	$\exp(-n\,C(p_0, p_1))$, $C := -\min_\lambda \ln \int p_0^\lambda p_1^{1-\lambda}$	None
Wald's martingale (SPRT)	when a sequential test may safely stop	a threshold crossing controls the error rate exactly, at any data-dependent stopping time	Fig. 3
Delta method	how biased and how noisy $\hat{L}_n$ is as an estimator of L_∞	bias $O(1/n)$, variance $O(1/n)$, both from a second-order Taylor expansion	Fig. 4
Fisher / Cramér-Rao	how precisely L's maximizer $\hat{\theta}$ can ever localize θ	$\mathrm{Var}(\hat{\theta}) \geq 1/(nI(\theta))$, reached asymptotically by the MLE	None

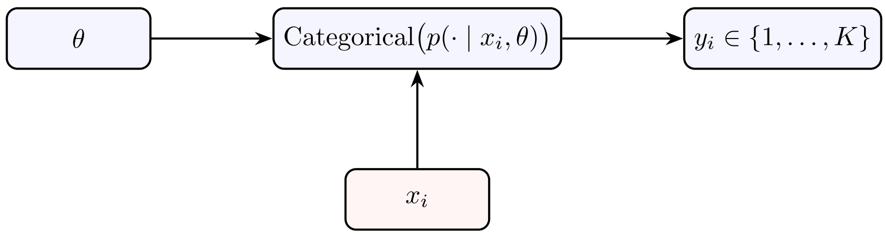
Table 2: The five asymptotic results Section 3 derives for the same quantity, L or the process built from it, side by side. Two have no dedicated figure. Chernoff information specializes the same large-deviations machinery as Cramér's row to a two-hypothesis decision rather than a single tail probability; the Fisher / Cramér-Rao row is a statement about $\hat{\theta}$'s own precision rather than a quantity with a natural curve to plot.

4 Two worked instantiations

The same construction, an exponentiated average log-density, lands on the reciprocal of a familiar error measure in both of the settings machine learning most often fits: a categorical classification model and a regression model under Gaussian noise.

4.1 Classification

A classifier's generative model is the same graphical model as Section 1's, specialized to a categorical outcome: x_i and θ set a probability vector over K classes; y_i is drawn from it.



Same reading as Section 1's diagram: x_i and θ feed the categorical law, which draws y_i from among K classes.

If Y is the true class and a model predicts $p(Y \mid X)$, the *cross-entropy* $\mathrm{CE} := -\mathbb{E}_{(X,Y)}[\ln p(Y \mid X)]$ measures, in nats, how surprised the model is on average by the true class; it is the quantity gradient descent fits in essentially every modern classifier. Exponentiated and sign-flipped, $\exp(\mathbb{E}_{(X,Y)}[\ln p(Y \mid X)]) = \exp(-\mathrm{CE})$ becomes exactly the geometric-mean probability the model assigns to the correct class, a reading no raw cross-entropy in nats offers directly. A cross-entropy of 0.693 nats, $\ln 2$, becomes $\exp(-0.693) \approx 0.5$: a model about as confident, on the correct class, as a coin flip. Equivalently, $\exp(\mathrm{CE})$ is the *perplexity* the model assigns to the true class, the effective number of classes it is choosing among as far as its confidence is concerned; a perplexity near 1 is a confident, well-calibrated model, a perplexity near K (the class count) is no better than guessing uniformly.

Take the $p = (0.5, 0.3, 0.15, 0.05)$ example Figure 1 already plotted, now read as one model's predicted probabilities for one example, class 1 the true label. The model's cross-entropy on that single example is $\mathrm{CE} = -\ln p_1 = -\ln 0.5 \approx 0.693$ nats, exactly the coin-flip case above: the model assigns the true class only even odds, even though it is the single most likely class under its own prediction. Averaged over many examples rather than read off one, the same $\exp(-\mathrm{CE})$ becomes the geometric-mean probability the model assigns to whichever class turns out true each time. Figure 1's right-hand panel is exactly that averaging machine turned into a dial: $\beta = 1$ is the ordinary, untilted cross-entropy read above, $\beta \to 0$ softens every prediction toward uniform confidence over all K classes regardless of the model's own opinion, while $\beta \to \infty$ hardens every prediction toward its single most likely class, a temperature knob turning a probabilistic classifier into something closer to a bare $\arg\max$ as it cools.

Figure 5 takes this from one hand-picked example to a genuine simulation: 45 points drawn from 3 overlapping classes, each assigned a probability for its own true class by a simple distance-to-centroid softmax model. Some points sit deep inside their own class's territory and score high; others sit near a boundary between two classes and score low, the model genuinely unsure which label is right. L is the geometric mean of all 45 of those individual probabilities, not an arbitrary summary of them: the right-hand panel sorts them low to high and marks exactly where that geometric mean falls, visibly closer to the crowd of low scores than an arithmetic mean would sit, the same pull toward the worst cases Jensen's inequality already predicted in Section 2.

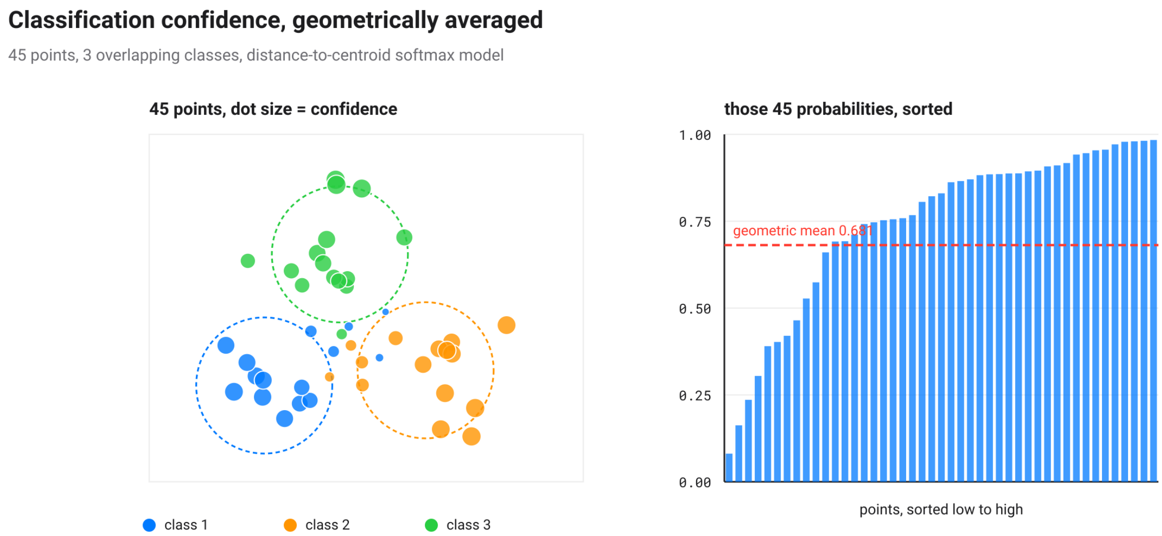


Figure 5: Left: 45 simulated examples from 3 overlapping classes (dashed rings mark the generating centroids), dot size showing the model's probability for each example's own true class, small near a class boundary, large deep inside a class's territory. Right: those same 45 probabilities sorted from low to high, with their geometric mean, this note's L, marked.

A single scalar, though, whether CE, $\exp(-\text{CE})$, or a perplexity, only ever answers "how confident, on average," never "how often actually right." A model can be confidently wrong, assigning high probability to an incorrect class. CE penalizes exactly that overconfidence more harshly than a small, honest uncertainty would be penalized, the same asymmetry a proper scoring rule is built to have, a rule constructed so that reporting your honest belief is always the best strategy: guessing 99% on a class that turns out false costs far more in nats than guessing 60% ever would. That asymmetry is why cross-entropy, rather than raw classification accuracy, is what gradient descent fits.

On real data: the Palmer Archipelago penguins. The simulated points above set the vocabulary; a real dataset, fit by real code, puts that vocabulary to work. The Palmer Archipelago, in Antarctica, is home to three species of *Pygoscelis* penguin, Adélie, chinstrap and gentoo, measured in the field by the ecological survey of Gorman et al. (2014): 344 birds, each with four real body measurements (bill length, bill depth, flipper length, body mass). It has become a modern teaching classic, preferred over older alternatives precisely because the three species separate cleanly without any sleight of hand.

A multinomial softmax classifier, exactly this section's own generative model with $K = 3$, is trained on those four standardized measurements by gradient descent on cross-entropy: 233 penguins for training, 100 held out for validation, never seen while the weights are fit. No library solver does the work; every iteration in Figure 6 is a genuine gradient-descent step, so the curves plotted are the real training curves, not something reconstructed after the fact. The model reaches 100% validation accuracy, with a final cross-entropy $\text{CE} \approx 0.024$ nats, a perplexity $\text{PPL} = \exp(\text{CE}) \approx 1.024$ (the model hesitates, on average, barely more than between one certain species) and $Q_{\text{classif}} \approx 0.978$.

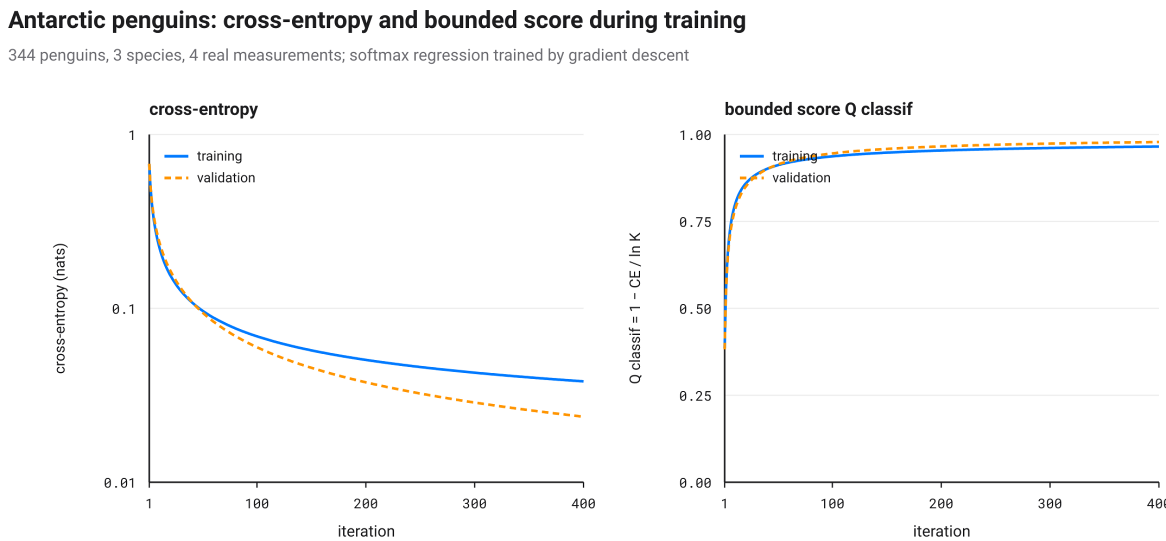
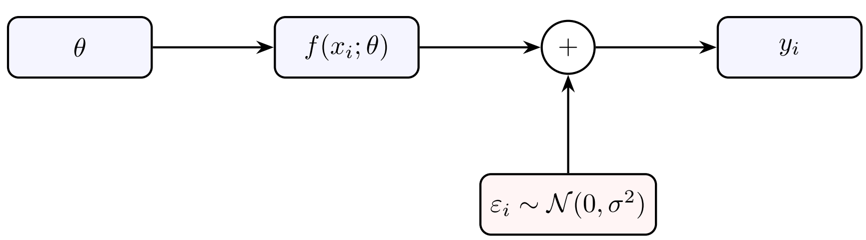


Figure 6: Training a multinomial softmax classifier on real measurements from 344 penguins of three species (Gorman et al., 2014) (233 for training, 100 for validation). Left: cross-entropy, logarithmic scale, decaying every gradient-descent iteration on both sets. Right: Section 5's bounded score Q_{classif}, climbing toward 1 as a mirror image of the cross-entropy. Validation ends up very slightly below training in the left panel: on a dataset this small, that is not overfitting but the same kind of sampling noise Section 3 already quantified for $\hat{L}_n$ itself, a variation from one validation draw to another rather than a property of the model.

4.2 Regression under Gaussian noise

Let $y_i = f(x_i;\theta) + \varepsilon_i$ with ε_i i.i.d. $\mathcal{N}(0,\sigma^2)$, f any regression function and θ its parameters, the generic conditional density of Section 1 specialized to one particular family:



Same reading again: θ sets the function f, whose output $f(x_i;\theta)$ is perturbed by Gaussian noise ε_i to produce y_i.

$$p(y_i \mid x_i, \theta) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{(y_i - f(x_i;\theta))^2}{2\sigma^2}\right) \tag{1}$$

so that, writing $\mathrm{MSE}(\theta) := \frac{1}{n}\sum_i (y_i - f(x_i;\theta))^2$, the mean-squared error already familiar from any least-squares fit,

$$L(\theta, \sigma^2) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{\mathrm{MSE}(\theta)}{2\sigma^2}\right). \tag{2}$$

Figure 7 draws this density literally rather than only in symbols: a small bell curve at a few chosen x positions, each one the conditional density $p(\cdot \mid x_i, \theta)$ turned on its side, its width set by σ and its center riding along the fitted line. The marked dot on each bell is that curve's own height at the point actually observed, $p(y_i \mid x_i, \theta)$, tall where the fit lands close to the data, short where it does not. L is nothing more than the geometric mean of those heights taken over every observation, not only the four marked ones, the exact continuous counterpart of Figure 5's sorted bars: a curve fit closely almost everywhere posts mostly tall heights and a large L, a curve that wanders posts some short ones and pulls the geometric mean down, one bad miss doing more damage to a geometric mean than to an arithmetic one, the same asymmetry Section 2's Jensen bound already flagged.

At the maximum-likelihood variance $\hat{\sigma}^2 = \mathrm{MSE}(\theta)$,

$$\exp(\mathbb{E}[\ln p]) = \frac{1}{\sqrt{2\pi e}\,\sqrt{\mathrm{MSE}(\theta)}}, \tag{3}$$

the reciprocal of the root-mean-square error up to a universal constant $\sqrt{2\pi e}$, the exact continuous counterpart of Section 4.1's reciprocal-perplexity reading: in both cases $\exp(\mathbb{E}[\ln p])$ is the reciprocal of an error measure native to the prediction task, perplexity for a categorical one, root-mean-square error for a continuous one. This is no coincidence. Writing $\ell := \ln L$ and, following the convention information criteria use, $\mathrm{PPL} := \exp(-2\ell)$, the identity $\exp(\mathbb{E}[\ln p]) = 1/\sqrt{\mathrm{PPL}}$ holds in general, Gaussian or not, a direct consequence of the two definitions rather than a property specific to either worked case. `elbow-helper`'s own `ELBOW-en.tex` takes this instantiation further: f there ranges over single lines, broken lines and k-segment piecewise lines. PPL is turned into a genuinely bounded, worst-case-anchored score rather than left as a bare reciprocal-RMSE reading.

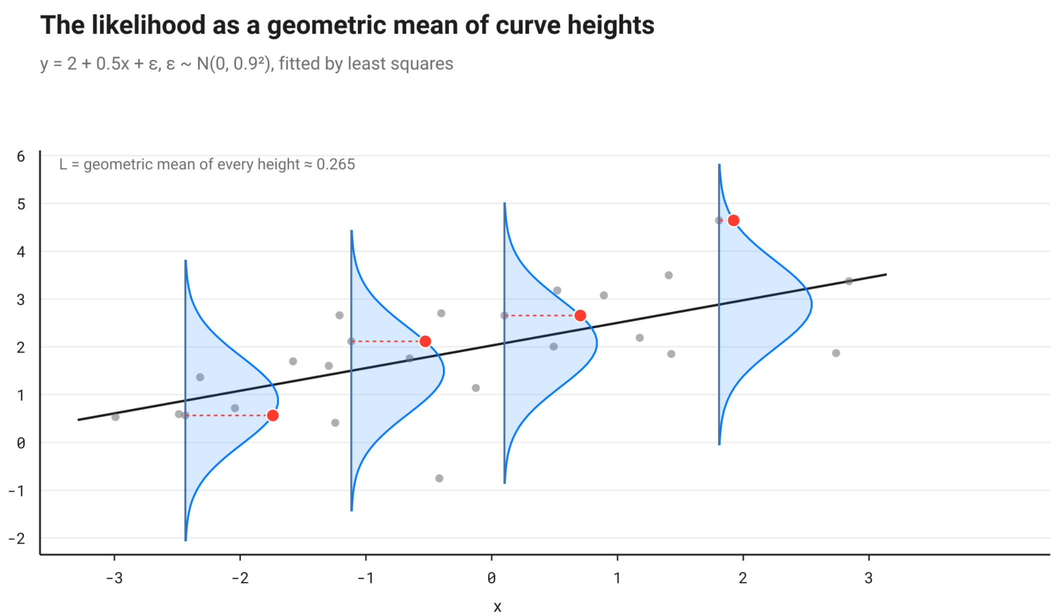


Figure 7: Synthetic regression data, $y = 2 + 0.5x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.9^2)$, with the fitted line and four conditional-density curves $p(\cdot \mid x, \theta)$ drawn at selected x positions. Each dot marks that curve's own height at the value actually observed there. L is the geometric mean of every such height, not only the four marked: high where the fit lands close to the data, low where a point falls far from the curve's center.

On real data: California housing. The same move repeats on a real regression dataset, larger and noisier than the synthetic line above. The California housing dataset (Pace and Barry, 1997), drawn from the 1990 census, gives 20,640 census blocks described by eight real variables (median block income, median housing age, average rooms and bedrooms per household, population, average occupancy, latitude, longitude) and the target to predict, the block's median house value. A linear model, $f(x_i;\theta) = \theta^\top x_i$ with x_i now an eight-variable standardized vector rather than a single number, generalizes this section's own $f(x;\theta)$ directly, without changing the construction at all; 16,512 blocks for training, 4,128 held out for validation, a model trained by gradient descent on mean-squared error rather than fit in one closed-form least-squares step, so the iteration curves are, here too, real.

The bounded score Q_{reg} Section 5 builds needs a worst case to anchor against; this experiment picks a simpler one than the hardest single point that same section will pull from the data, so it can stand on its own: the $\text{MSE}_{\text{worst}}$ plotted here is the error of a model that uses none of the eight variables and predicts the training mean everywhere, the regression counterpart of the uniform guess that anchors Q_{classif}. On validation, that naive model reaches $\text{MSE}_{\text{worst}} \approx 1.31$ (hundred-thousand dollars squared); the trained linear model finishes at $\text{MSE} \approx 0.556$, $\text{RMSE} \approx 0.746$ (about \$74,600 of typical error on a median value ranging from \$15,000 to \$500,000), and $Q_{\text{reg}} \approx 0.576$. That score, well short of 1, is not a sign the code failed but an honest reading: eight linear variables cannot capture the full structure of a real housing market. Q_{reg} says so in a single number rather than hiding it behind a mean-squared error whose scale alone says nothing.

Figure 8: Training a linear regression on the 8 real variables of the California housing census data (Pace and Barry, 1997) (16,512 blocks for training, 4,128 for validation). Left: mean-squared error at every gradient-descent iteration. Right: Section 5's bounded score Q_{reg}, anchored here on the error of a covariate-free model (the training mean predicted everywhere), rising as a mirror image of the mean-squared error and plateauing around 0.58: a linear model's real limit on real data, not a bug.

That linear ceiling invites a natural next question: how much of the gap comes from the eight variables themselves and how much from the straight-line shape forced on them? The same likelihood, the same standardized split, and the same gradient descent carry over unchanged; only $f(x; \theta)$ grows more expressive. A multilayer perceptron answers it: one hidden layer of 30 ReLU (rectified linear unit, the simple $\max(0, x)$ nonlinearity used in most modern hidden layers) units on He-initialized weights (He et al., 2015), a random initialization scaled to the layer size so signal and gradient neither explode nor vanish through the ReLU layers, feeding a single linear output unit, the exact architecture Géron's textbook (Géron, 2019) uses for this dataset. Trained here by hand-written forward and backward passes, not a library call, on the identical 16,512/4,128 split, it reaches on validation MSE ≈ 0.343, RMSE ≈ 0.586 (about \$58,600 of typical error, some \$16,000 less than the linear model's \$74,600), and $Q_{\text{reg}} \approx 0.738$. The gap was never in the likelihood construction; it was in the straight line.

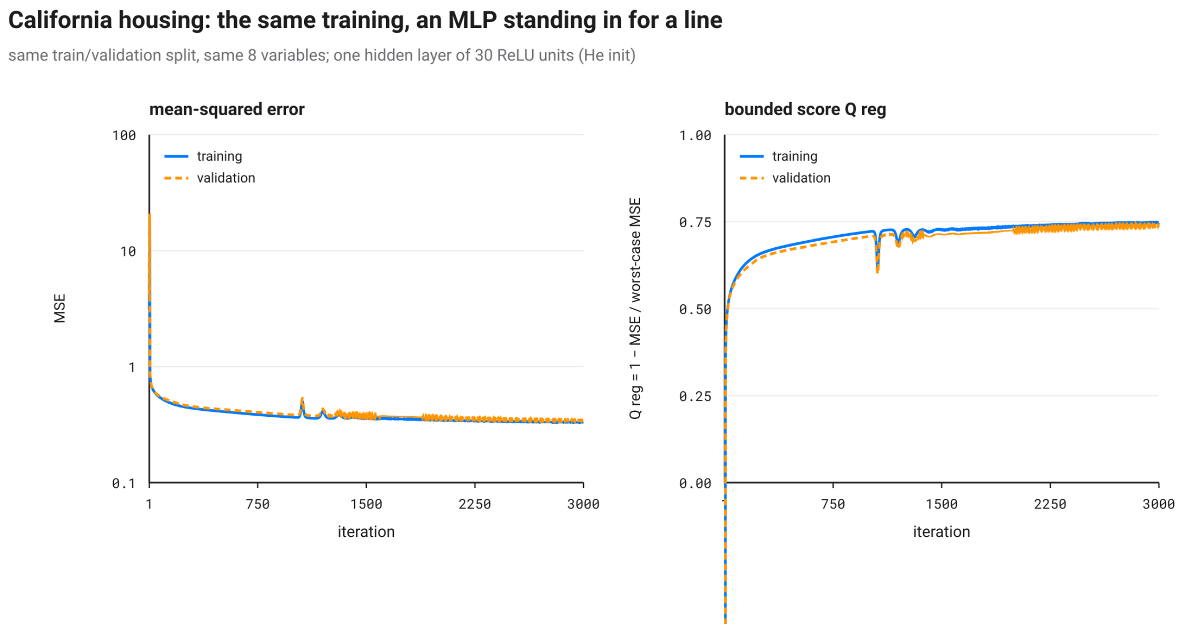


Figure 9: The same California housing data (Pace and Barry, 1997) and the same 16,512/4,128 split as Figure 8, now fit by a one-hidden-layer MLP (30 ReLU units, He init (He et al., 2015), Géron's architecture (Géron, 2019)) instead of a line. Left: mean-squared error, log-scaled to show both the first gradient step's overshoot from a random initialization and the true convergence beneath it; the small wobble near iteration 1,000 is a real, minor instability where the loss surface's non-smoothness at the ReLU kinks briefly shows through, not a plotting artifact. Right: Q_{reg} climbing past the linear model's 0.58 ceiling to settle near 0.74.

5 Bounding the likelihood: a practical construction

Every quantity this note built, CE, ℓ, PPL, is a rescaling of the same underlying number. Every one of those rescalings shares the same first problem: taken raw, it has no ceiling, so no single threshold on it means the same thing across two different fits. Fixing that takes anchoring against a real, achievable worst case rather than an arbitrary or unreachable one, a move classification needs once and regression needs twice over, for a reason worth making precise rather than waved at.

Classification's $\mathrm{CE}(\theta) \geq 0$ already has a clean floor: zero exactly at a perfectly confident, correct prediction, nothing special has to be done to reach it. Section 2's temperature sweep already named the natural ceiling to anchor against. At $\beta \to 0$, hot and maximally agitated, any model's confidence flattens to uniform over the K classes regardless of what it actually believed. A model that predicts uniformly earns exactly $\mathrm{CE} = \ln K$ on any true class whatsoever, the one cross-entropy value that does not depend on which class turns out right. That is not an arbitrary reference: it is what refusing to use any of the model's own information costs, in nats, no matter the label, so

$$Q_{\mathrm{classif}}(\theta) := 1 - \frac{\mathrm{CE}(\theta)}{\ln K}$$

reads 1 for a perfectly confident, correct model, 0 for one no better than uniform guessing, and negative for one actively misleading, confidently wrong more often than confidently right, a signal this note sees no reason to hide by clipping it away.

Regression's own log-likelihood has no such floor. That is the second problem, distinct from the ceiling both cases share. It traces back to the density-versus-probability distinction drawn in Section 1: a continuous density can exceed 1, so profiling σ^2 at its best-supported value $\hat{\sigma}^2 = \mathrm{MSE}(\theta)$ sends $\ell(\theta, \hat{\sigma}^2)$ to $+\infty$, not to a clean 0, exactly at a perfect fit, the classical likelihood degeneracy behind BIC's whole reason to exist. Exponentiating repairs that first: $\mathrm{PPL}(\theta) := \exp(-2\ell(\theta, \hat{\sigma}^2)) = 2\pi e \cdot \mathrm{MSE}(\theta) \geq 0$, zero exactly at a perfect fit, still unbounded above but now starting from a floor worth building on. `elbow-helper`'s own `ELBOW-en.tex` supplies the second move, the ceiling: not a formula but a genuine worst case drawn from the data itself, the single *observed* point y_{bad} hardest to predict everything else from, giving $\mathrm{MSE}_{\mathrm{worst}}$ the exact role $\ln K$ plays above,

$$Q_{\mathrm{reg}}(\theta) := 1 - \frac{\mathrm{MSE}(\theta)}{\mathrm{MSE}_{\mathrm{worst}}} = 1 - \frac{\mathrm{PPL}(\theta)}{\mathrm{PPL}_{\mathrm{worst}}},$$

the two ratios equal because PPL and MSE carry the same constant $2\pi e$ on both the numerator and the denominator.

Two different worst cases answer the same question in the only two ways this note's own generative model allows. A K-category classifier has a fixed, label-independent alternative built into its own structure (uniform guessing) and never has to look at the data to find one. A continuous regressor has no such built-in alternative: nothing about the model family alone says what a bad prediction looks like, so the worst case has to be borrowed from the very sample the model is trying to fit. Once that asymmetry is priced in rather than papered over, CE and MSE, numbers that only ever meant something next to another run of the same model on the same data, become Q_{classif} and Q_{reg}, numbers that mean the same thing on any dataset, of any size, in either language this note is written in.

The parallel, laid out once. Every quantity in Section 4 has a twin on the other side of the classification/regression divide; Table 3 lines them up so the correspondence is visible at a glance rather than assembled from five separate paragraphs.

Role	Classification	Regression
Raw scale (nats)	$\mathrm{CE} := -\mathbb{E}[\ln p(Y \mid X)]$	$\ell := \ln L = \mathbb{E}[\ln p]$
Exponentiated	$\exp(-\mathrm{CE})$, geometric-mean confidence	$\exp(\ell) = \dfrac{1}{\sqrt{2\pi e}\sqrt{\mathrm{MSE}}}$
Reciprocal error reading	perplexity, $\mathrm{PPL} = \exp(\mathrm{CE})$	RMSE, up to the constant $\sqrt{2\pi e}$
Worst-case anchor	$\ln K$, uniform guessing over K classes	$\mathrm{MSE}_{\mathrm{worst}}$, the hardest observed point
Bounded score	$Q_{\mathrm{classif}} = 1 - \mathrm{CE}/\ln K$	$Q_{\mathrm{reg}} = 1 - \mathrm{MSE}/\mathrm{MSE}_{\mathrm{worst}}$

Table 3: The same construction, an exponentiated average log-density, read through a categorical model (left) and a Gaussian model (right). Every row is one instance of the identity $\exp(\mathbb{E}[\ln p]) = 1/\sqrt{\mathrm{PPL}}$ from Section 4.2, not two unrelated coincidences that happen to look similar.

6 Going further: a Gaussian process over functions

Intuition. Every model this note has fit picks one function $f(x; \theta)$ and reports how likely the data looks under it. A Gaussian process keeps a whole family of plausible functions on hand at once, weighted by how well each one fits. It predicts by averaging over that family rather than committing to a single curve. That sounds like it should need new machinery; it does not. The same Gaussian log-likelihood Section 4.2 built reappears unchanged, applied to a vector of function values instead of one scalar prediction.

The friction point. A distribution over an object as large as "a function" sounds like it needs new mathematics: a function has, in principle, infinitely many outputs, one per possible input x. A Gaussian process sidesteps this by never writing down that infinite-dimensional object directly. It only ever describes finitely many outputs at a time. Pick any finite set of locations $x_1, \ldots, x_n$; the process declares the corresponding function values $f(x_1), \ldots, f(x_n)$ jointly Gaussian, mean zero by convention, covariance $K_{ij} := k(x_i, x_j)$ set by a *kernel* function k. Ask about a different, larger finite set of points and the same rule gives a consistent joint Gaussian on that larger set too; that consistency is the entire content of the word "process" here (Rasmussen and Williams, 2006). In practice a Gaussian process is used through exactly two operations, choose a kernel, then condition that one big joint Gaussian on the observed data, never anything more exotic.

The kernel. The squared-exponential kernel,

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right),$$

has two knobs. σ_f^2 sets the vertical scale of typical variation. ℓ, the *lengthscale*, sets the horizontal one: how far apart two x-values must sit before the kernel calls them nearly uncorrelated, which is exactly how quickly the process is allowed to change. Points close together in x get a covariance near σ_f^2, nearly identical function values; points many lengthscales apart get a covariance near 0, free to differ.

Fitting with noise, and the return of the familiar likelihood. Real observations carry noise: $y_i = f(x_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$ i.i.d., the exact generative picture of Section 4.2, applied now to the whole vector $y := (y_1, \ldots, y_n)$ at once rather than one y_i at a time. Under the Gaussian process prior, y itself is jointly Gaussian, $y \sim \mathcal{N}(0, K + \sigma_n^2 I)$, so

$$\ln p(y \mid X, \theta) = -\frac{1}{2} y^\top (K + \sigma_n^2 I)^{-1} y - \frac{1}{2} \ln|K + \sigma_n^2 I| - \frac{n}{2} \ln(2\pi), \tag{4}$$

the *log marginal likelihood*: marginal because it is what is left of the joint density $p(y, f \mid X)$ after integrating f out entirely, not just plugging in the one f that happens to fit best. Equation (4) is not new mathematics; it is $\ln L_{\text{raw}}(\theta)$ from Section 3, generalized from a diagonal covariance to a full one. Setting $\sigma_f^2 \to 0$ collapses K to the zero matrix, $K + \sigma_n^2 I$ to $\sigma_n^2 I$, and Equation (4) exactly to $n\,\ell(\theta, \sigma_n^2)$, n copies of Section 4.2's own per-observation log-likelihood summed rather than averaged, the same sum-versus-mean relation Section 3 built for exactly this reason.

Prediction. Conditioning a joint Gaussian on part of its own coordinates is a standard, fully worked-out linear-algebra identity (Rasmussen and Williams, 2006): writing K_* for the covariance between the n training points and m new test points and K_{**} for the covariance among the test points themselves, the posterior over function values at the test points is again Gaussian,

$$\mu_* = K_*^\top (K + \sigma_n^2 I)^{-1} y, \qquad \Sigma_* = K_{**} - K_*^\top (K + \sigma_n^2 I)^{-1} K_*,$$

mean and covariance both computable from the same Cholesky factor of $K + \sigma_n^2 I$ that Equation (4) already needs, the standard way to factor a covariance matrix into a triangular square root so that a linear solve and a log-determinant can both be computed stably and cheaply. Intuitively μ_* pulls each test point's prediction toward nearby training values, weighted by the kernel. Σ_* shrinks near training data and grows in gaps the data never covered: the model's own honest account of where it is guessing.

Choosing the kernel's own hyperparameters. Hyperparameters here means knobs that shape the model before it ever sees data, as opposed to the parameters θ fit directly to it everywhere else in this note. The same principle every earlier section used, maximize the likelihood, applies again, except now what is being tuned is not a curve's slope and intercept but the *shape* of the covariance itself: how smooth the model is even allowed to be before it sees a single data point. Maximizing Equation (4) over $\theta = (\ell, \sigma_f^2, \sigma_n^2)$, the symbol reused here for the covariance's own three knobs rather than Section 4.2's curve parameters, is called type-II maximum likelihood, or empirical Bayes, a name that is literal: in a fully Bayesian treatment the hyperparameters would themselves carry a prior, but here they are simply point-estimated from the very data they will go on to explain. Type-I would maximize over the function values f themselves at fixed hyperparameters; type-II integrates f out first (Equation (4) already reflects that integration) and maximizes over what is left.

On real data: California housing, one more time. A 50-block subsample of the same California housing dataset (Pace and Barry, 1997) Section 4.2 already used, one feature, median block income, against the target, median house value, kept to one dimension and a small n so the exact $O(n^3)$ Cholesky solve stays cheap and a plotted uncertainty band stays legible. No GP library: the kernel, the Cholesky solve, Equation (4) and its grid-search maximization are plain numpy, the same from-scratch discipline as every other model in this note. σ_f^2 is fixed to the sample variance of y, a standard, data-driven default for the signal scale; ℓ and σ_n^2 are chosen by a grid search maximizing Equation (4), landing on $\ell^* \approx 3.13$ (standardized income units), $\sigma_n^2 \approx 0.49$, a raw log marginal likelihood of about -62.90 summed over the $n = 50$ points: exactly the unbounded, sample-size-dependent nats count Section 5 built Q_{reg} to stop reporting raw. The same fix applies here without a single new principle: average per observation rather than sum, $\ell_{\text{gp}}(\theta) := \frac{1}{n} \ln p(y \mid X, \theta)$, exponentiate into a perplexity, $\text{PPL}_{\text{gp}}(\theta) := \exp(-2\ell_{\text{gp}}(\theta))$, and normalize against this 50-block sample's own worst case, computed by the identical y_{bad} recipe Section 5 borrows from `ELBOW-en.tex`,

$$Q_{\text{gp}}(\theta) := 1 - \frac{\text{PPL}_{\text{gp}}(\theta)}{\text{PPL}_{\text{worst}}}, \qquad \text{PPL}_{\text{worst}} := 2\pi e \cdot \text{MSE}_{\text{worst}},$$

landing on $\text{MSE}_{\text{worst}} \approx 9.07$, $\text{PPL}_{\text{worst}} \approx 154.88$ for this particular 50 blocks, and $Q_{\text{gp}}(\theta^*) \approx 0.92$ at the chosen hyperparameters, a number that means the same thing next to Q_{reg} from Section 5 rather than only next to another lengthscale on the same curve.

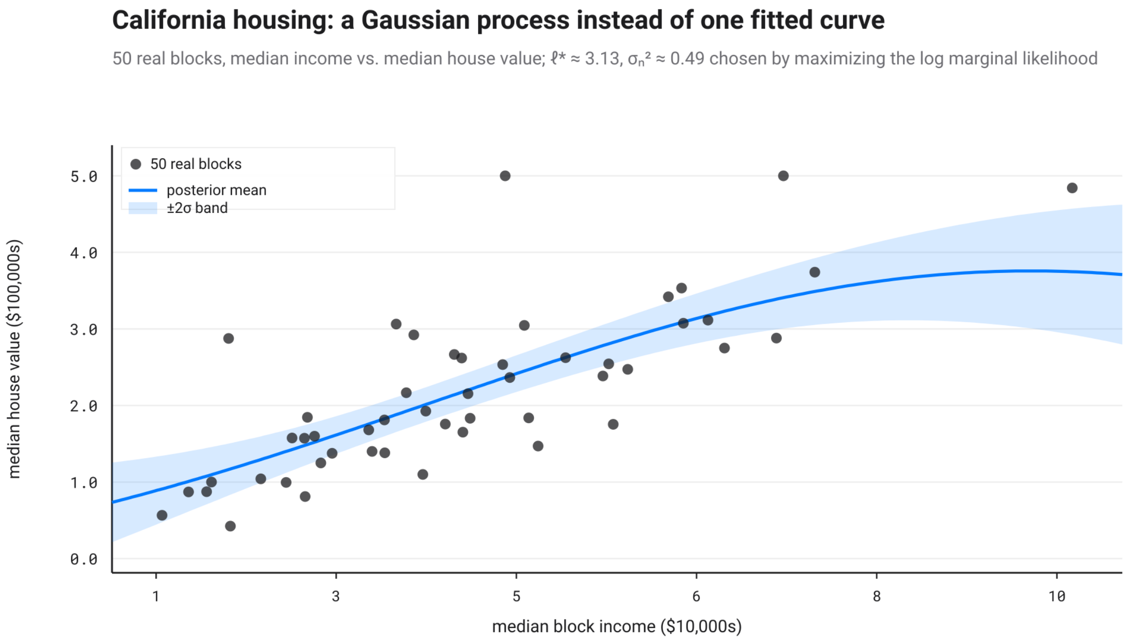


Figure 10: The fitted Gaussian process on 50 real census blocks (Pace and Barry, 1997): posterior mean (solid) and $\pm 2\sigma$ band (shaded) at every income level, not only the 50 observed ones. The band narrows where blocks are dense and widens past roughly $70,000 of median income, where real observations thin out and the model is honestly less sure what a plausible house value looks like.

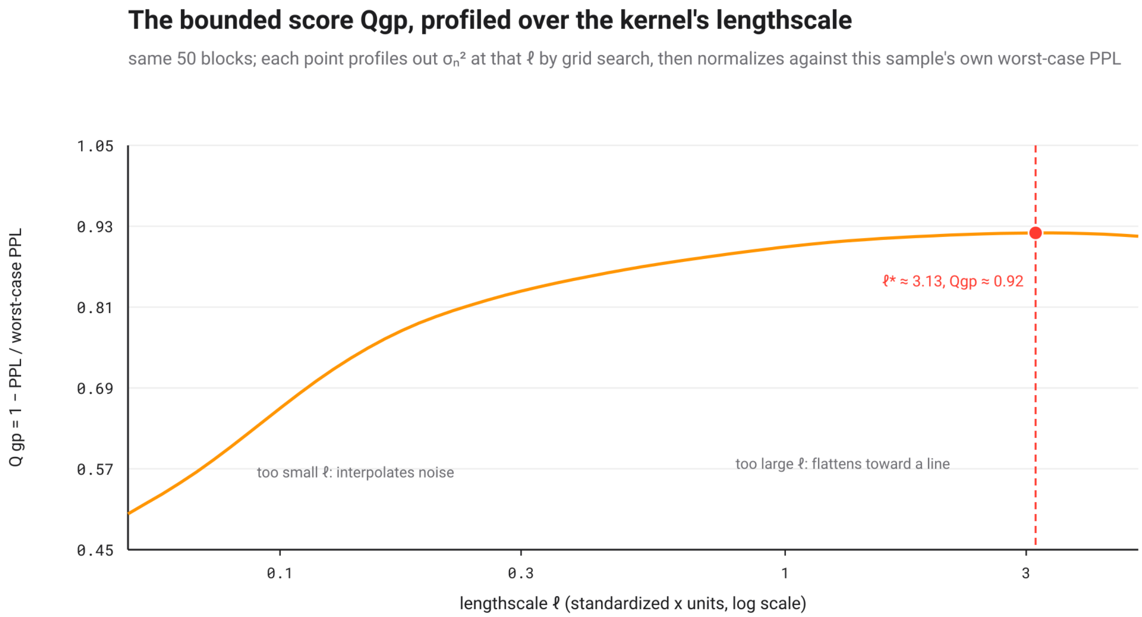


Figure 11: The same 50 blocks' bounded score Q_{gp}, not the raw log marginal likelihood, profiled over the lengthscale ℓ: at each ℓ on the horizontal axis, σ_n^2 is re-optimized by the same grid search over Equation (4), then the resulting best score at that ℓ is converted to Q_{gp} exactly as defined in the text, so the curve stays on the same $[0, 1]$-anchored scale as every other bounded score this note plots, comparable across runs rather than legible only next to its own neighboring points. Too small an ℓ lets the process interpolate every training point, including its noise; Q_{gp} punishes that with a low score. Too large an ℓ flattens the fit toward a single straight line and loses real signal the data actually supports. The interior maximum at $\ell^* \approx 3.13$, $Q_{\text{gp}} \approx 0.92$, is the same tradeoff Equation (4) was already resolving; normalizing only rescales the vertical axis, it does not move where the maximum sits, with no held-out validation split required to find it (Rasmussen and Williams, 2006).

Two of the 50 real blocks in this experiment happen to sit at \$500,000, this dataset's own census-reporting cap rather than a true ceiling on house values; the Gaussian process was not told this. It treats those two points as any other noisy observation. Section 5's bounded scores were built for exactly this kind of imperfection: real data rarely arrives clean; a likelihood-based model, whether one straight line, an MLP, or a whole distribution over curves, only ever reports how well it explains the data actually given to it.

7 Conclusion

Every quantity this note touched, cross-entropy, perplexity, regression log-likelihood, the Gaussian process's marginal likelihood, turned out to be one identity wearing a different name: $L := \exp(\mathbb{E}[\ln p])$, an exponentiated average log-density rather than the raw product most references open with. Reading likelihood this way is what let Section 4's two worked models, a softmax classifier and a Gaussian regressor, land on the same construction from opposite sides of the discrete/continuous divide. It is what let Section 5 turn two model-specific, unbounded numbers into Q_{classif} and Q_{reg}, comparable across datasets because each is anchored to a real worst case rather than an arbitrary one. It is what let Section 6 extend the whole argument from one predicted value to a full distribution over functions without introducing a single new principle, only a larger covariance to invert.

None of this note's four experiments, the penguin classifier, the housing line, the housing MLP, the housing Gaussian process, needed a library built around likelihood to make that identity concrete: numpy, one gradient loop or one Cholesky solve, and the formula itself were enough in every case. That is the claim this note set out to defend: not that likelihood demands separate machinery for classification, regression or Gaussian processes, but that it demands none, once treated as an average rather than a product. `elbow-helper`'s own `ELBOW-en.tex` takes that foundation as settled and builds its Gaussian model-selection criteria directly on top of it; this note's job ends where that one begins.

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